

802.11b

The IEEE expanded on 802.11 in July 1999, creating the 802.11b specification. 802.11b supports bandwidths of up to 11 Mbps, comparable to Ethernet.

802.11b uses the same radio signalling frequency—2.4 GHz—as the original 802.11 standard. Being an unregulated frequency, 802.11b devices can incur interference from microwave ovens, cordless phones, and other appliances using the 2.4 GHz range.

802.11a

When 802.11b was developed, the IEEE created a second extension to the 802.11 standard, called 802.11a. Due to its higher cost, 802.11a fits in well with the business market, whereas 802.11b serves the home market better.

802.11a supports a maximum bandwidth of up to 54 Mbps and signals in a regulated 5 GHz range. The higher frequency means the range of 802.11a is more limited. It also means 802.11a signals have more difficulty penetrating walls and other obstructions. Because 802.11a and 802.11b utilise different frequencies, the two technologies are incompatible with each other.

802.11g

In 2002 and 2003, WLAN products supporting a new standard called 802.11g began to appear. 802.11g attempts to combine the best of both 802.11a and 802.11b. 802.11g supports bandwidths of up to 54 Mbps, and it uses the 2.4 GHz frequency for greater range. 802.11g is backwards compatible with 802.11b.

Trends

Anand Mehta, manager, new business areas development, D-Link India, says, "Wireless (WLAN) is primarily a platform for data transmission without use of a solid state media. Here one is essentially referring to IEEE 802.11 (a, b, g, i, etc.) standards.

"The WLAN market is to a large extent driven by the market for laptops and the penetration rate of laptops vs. desktops would define the extent to which WLANs are being deployed. Here in India, the penetration of PCs into homes is much lower compared to its presence in Business Environments. Simultaneously, it is most inconceivable to imagine that we would see a large number of homes having two PCs in India for some time. Hence, here, wireless deployments are mainly in enterprises.

"Other drivers for WLANs are scenarios like rented office space premises that may need to be shifted, or Cyber-Cafes, or deployments in the hospitality industry. Besides the above, there are a few niche applica-

tions such as RFID checks, industrial applications requiring the noting down of reading from various sensors, and real-time status monitoring in hospitals.

"VoIP, when it first came, was heralded as a system that would reduce the cost of communications drastically, and research organisations predicted/anticipated that all phones would be based on VoIP.

"This has happened only partially. True, VoIP has significantly helped in reduction of costs and thus prices, and enabled many smaller players to enter the normally very capital intensive area of telecom, yet on the other hand, one does not really find many telephone services being deployed completely on VoIP. One sees VoIP either being deployed by the carriers (on their back-end) or being used through dial-up IP phones or used within organisations for their internal communications on VPNs or Leased Circuit Networks.

"VoIP over Wireless is essentially a tool to enable seamless deployments of VoIP within the organisation. It is a solution offered today by quite a few vendors and is constantly improving. One could even have these services deployed in homes as the last mile, but here one would need to have broadband as the base service on which one would deploy VoIP though Wireless.

"I envision VoIP over WLAN in the Indian context to be deployed only by enterprises till one sees a service provider deploy broadband in a really big way, and then at a later stage, maybe a couple of years down the line, extend this to VoIP.

"The main market barriers for VoWLAN are existing network infrastructure and operating devices. These will have to be replaced with VoWLAN-compliant products, affecting the current projected revenues of mobile phone carriers and telephone operators. This makes it difficult to predict if telecom service providers will encourage or hamper the deployment for VoWLAN.

"WLAN will remain in use mostly for data transmission in the next two years. But usage of VoWLAN offshoots such as WiFi phones, PDAs or notebook PCs with built-in voice functions for home and office users will increase notably in 2006 (worldwide) and India should latch on in a year or two after this."

We asked Prashant Naik, a home PC enthusiast, about what he would like to see happening in the wireless arena in 2005. Here's what he had to say to us:

"I would like to have a wireless network in my house; I already have a Centrino notebook, and am planning to install an access point so that I can access cable Internet on by PC as well as my laptop; and my brother can do the same on his laptop.

"At present, the cost of an access point is a little high, at Rs 7,500. I'd like to buy one when it drops to around 2,500—the price of an external modem. I am expecting the price to fall this year as Centrino sales increase." ■

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